

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **4-Iodophenylhydrazine**  
 Product Description: **4-Iodophenylhydrazine**  
 Cat No. : 399050000; 399050010; 399050050  
 CAS No 13116-27-3  
 Molecular Formula C<sub>6</sub> H<sub>7</sub> I N<sub>2</sub>

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company** Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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 Main line: +60 3-5525 7888

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 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Acute oral toxicity                                | Category 3 (H301) |
| Acute dermal toxicity                              | Category 3 (H311) |
| Acute Inhalation Toxicity - Dusts and Mists        | Category 3 (H331) |
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319) |
| Skin Sensitization                                 | Category 1 (H317) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |

**Label Elements**

**Signal Word**
**Danger**

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## Hazard Statements

H317 - May cause an allergic skin reaction  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H301 + H311 + H331 - Toxic if swallowed, in contact with skin or if inhaled

## Precautionary Statements

### Prevention

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P311 - Call a POISON CENTER or doctor  
P330 - Rinse mouth  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                 | CAS No     | Weight % |
|---------------------------|------------|----------|
| 1-(4-Iodophenyl)hydrazine | 13116-27-3 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

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**Inhalation** Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.

**Self-Protection of the First Aider** Use personal protective equipment as required.

**Most important symptoms and effects, both acute and delayed**

May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>), Hydrogen iodide.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

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## Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

## Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.  
To maintain product quality: Store under an inert atmosphere, Keep refrigerated.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

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## Information on basic physical and chemical properties

### Appearance

### Physical State

Solid

### Odor

No information available

### Odor Threshold

No data available

### pH

No information available

### Melting Point/Range

101 - 102 °C / 213.8 - 215.6 °F

### Softening Point

No data available

### Boiling Point/Range

No information available

### Flash Point

No information available

**Method -** No information available

### Evaporation Rate

Not applicable

Solid

### Flammability (solid,gas)

No information available

### Explosion Limits

No data available

### Vapor Pressure

No data available

### Vapor Density

Not applicable

Solid

### Specific Gravity / Density

No data available

### Bulk Density

No data available

### Water Solubility

Insoluble

### Solubility in other solvents

No information available

## Partition Coefficient (n-octanol/water)

### Component

### log Pow

1-(4-Iodophenyl)hydrazine

3.017

### Autoignition Temperature

No data available

### Decomposition Temperature

No data available

### Viscosity

Not applicable

Solid

### Explosive Properties

No information available

### Oxidizing Properties

No information available

### Molecular Formula

C6 H7 I N2

### Molecular Weight

234.04

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Light sensitive. Air sensitive.

### Possibility of Hazardous Reactions

#### Hazardous Polymerization

Hazardous polymerization does not occur.

#### Hazardous Reactions

None under normal processing.

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## Conditions to Avoid

Incompatible products. Excess heat. Exposure to light. Exposure to air.

## Incompatible Materials

Strong oxidizing agents. Strong acids.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Hydrogen iodide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 3 |
| Dermal     | Category 3 |
| Inhalation | Category 3 |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

|             |                                         |
|-------------|-----------------------------------------|
| Respiratory | No data available                       |
| Skin        | Category 1                              |
|             | May cause sensitization by skin contact |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3  
Results / Target organs Respiratory system.

(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

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|                                                   |                                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                         |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.                                                    |

## SECTION 12: ECOLOGICAL INFORMATION

| <u>Ecotoxicity effects</u>                                                                                                                                                          | Do not empty into drains.                                                                                  |                               |                               |                           |       |                   |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|---------------------------|-------|-------------------|--|--|
| <u>Persistence and degradability</u>                                                                                                                                                |                                                                                                            |                               |                               |                           |       |                   |  |  |
| Persistence                                                                                                                                                                         | Persistence is unlikely.                                                                                   |                               |                               |                           |       |                   |  |  |
| <u>Bioaccumulative potential</u>                                                                                                                                                    | Bioaccumulation is unlikely                                                                                |                               |                               |                           |       |                   |  |  |
| <table><tr><th>Component</th><th>log Pow</th><th>Bioconcentration factor (BCF)</th></tr><tr><td>1-(4-Iodophenyl)hydrazine</td><td>3.017</td><td>No data available</td></tr></table> | Component                                                                                                  | log Pow                       | Bioconcentration factor (BCF) | 1-(4-Iodophenyl)hydrazine | 3.017 | No data available |  |  |
| Component                                                                                                                                                                           | log Pow                                                                                                    | Bioconcentration factor (BCF) |                               |                           |       |                   |  |  |
| 1-(4-Iodophenyl)hydrazine                                                                                                                                                           | 3.017                                                                                                      | No data available             |                               |                           |       |                   |  |  |
| <u>Mobility in soil</u>                                                                                                                                                             | Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility. |                               |                               |                           |       |                   |  |  |
| <u>Endocrine Disruptor Information</u>                                                                                                                                              | This product does not contain any known or suspected endocrine disruptors                                  |                               |                               |                           |       |                   |  |  |
| <u>Other adverse effects</u>                                                                                                                                                        | No information available                                                                                   |                               |                               |                           |       |                   |  |  |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste treatment methods</b>             |                                                                                                                                                                     |
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                           |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains                                         |

## SECTION 14: TRANSPORT INFORMATION

|                             |                              |
|-----------------------------|------------------------------|
| <b>IMDG/IMO</b>             |                              |
| <b>UN-No</b>                | UN2811                       |
| <b>Hazard Class</b>         | 6.1                          |
| <b>Packing Group</b>        | III                          |
| <b>Proper Shipping Name</b> | Toxic solid, organic, n.o.s. |

|                                |        |
|--------------------------------|--------|
| <b>Road and Rail Transport</b> |        |
| <b>UN-No</b>                   | UN2811 |
| <b>Hazard Class</b>            | 6.1    |

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|                             |                              |
|-----------------------------|------------------------------|
| <b>Packing Group</b>        | III                          |
| <b>Proper Shipping Name</b> | Toxic solid, organic, n.o.s. |

## IATA

|                             |                              |
|-----------------------------|------------------------------|
| <b>UN-No</b>                | UN2811                       |
| <b>Hazard Class</b>         | 6.1                          |
| <b>Packing Group</b>        | III                          |
| <b>Proper Shipping Name</b> | Toxic solid, organic, n.o.s. |

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

### National Regulations

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| <b>Persistent Organic Pollutant</b> | This product does not contain any known or suspected substance |
| <b>Ozone Depletion Potential</b>    | This product does not contain any known or suspected substance |

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS



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|                  |                 |
|------------------|-----------------|
| Revision Date    | 22-Mar-2025     |
| Revision Summary | Not applicable. |

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**